Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919588, or equivalent
- Cummins® Vehicle Air Plumbing Clean Care Kit, Part Number 4919588, or equivalent
- Exhaust seal installer, Part Number 4918709, or equivalent.

Additional Service Items

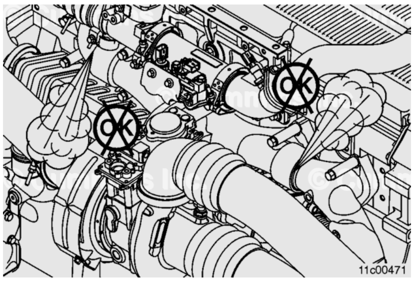
- No additional service items required.

Initial Check

Note : This engine family is equipped with exhaust seal rings at the slip joints.

Operate the engine and inspect the exhaust manifold for leaks at the slip joints.

If the exhaust manifold is leaking at the slip joints, it **must** be repaired.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

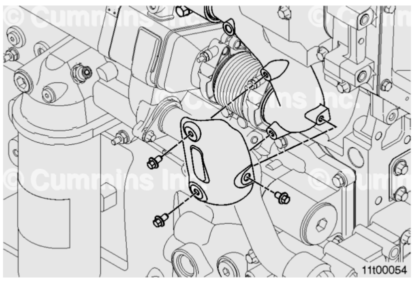
Note : Brush away any loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Remove the charge air cooler piping. See equipment manufacturer service information. Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, or equivalent, to cover the connection points.
- Remove the air intake pipe to the turbocharger. See equipment manufacturer service information. Use protective caps from the Cummins® Vehicle Air Plumbing Clean Care Kit, Part Number 4919588, or equivalent, to cover the connection points.
- Loosen the bellows connection between the exhaust manifold and exhaust gas recirculation (EGR) cooler.
- Remove the variable geometry turbocharger (VGT). Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-027.html)

Remove

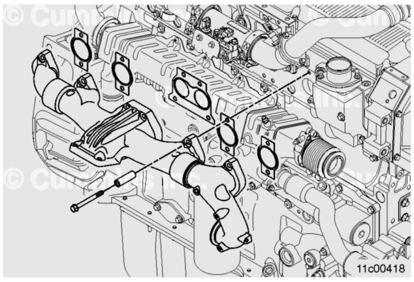
Remove the heat shield mounting capscrews from the exhaust manifold.

Remove the heat shield.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the two upper mounting capscrews and install two guide studs into the top center section mounting location.

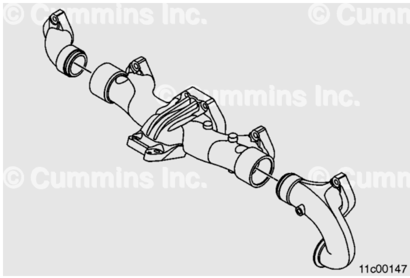
Remove the remaining mounting capscrews, spacers, and the exhaust manifold assembly.

Remove and discard the gaskets.

Cover open points on the exhaust manifold and cylinder head using heavy tape provided in the Air Handling Clean Care Kit, Part Number 4919588.

Disassemble

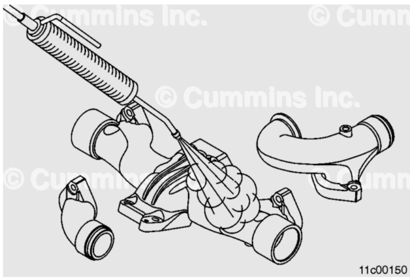
Disassemble the exhaust manifold end sections.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing, to reduce the possibility of personal injury when using a steam cleaner or high pressure water.



⚠ WARNING ⚠

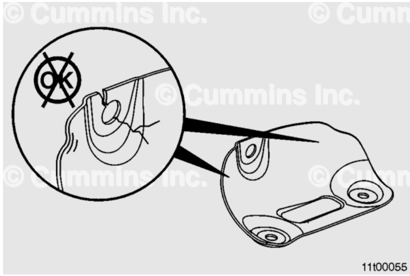
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the exhaust manifold sections and heat shield.

Dry with compressed air.

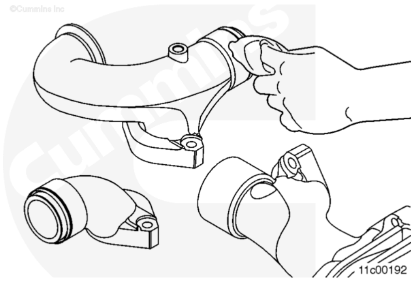
Inspect the heat shield for cracks or other damage.

Replace the heat shield if cracks or other damage is found



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

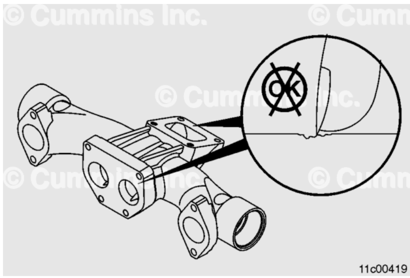


Use a soft cloth or a shop towel wet with a safety solvent to remove carbon deposits from the sealing surfaces.

Inspect the mounting surfaces for cleanliness and damage.

Inspect the exhaust manifold turbocharger mounting flange for warpage.

After cleaning, if visible gouges can be felt with a finger across the sealing bead area, it is **not** acceptable. Replace the exhaust manifold



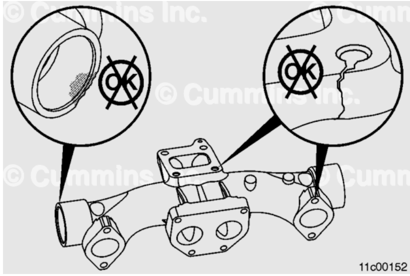
The exhaust manifold mounting surface **must** be flat to within 0.25 mm [0.010 in]. The turbocharger mounting flange **must** have a surface flatness of 0.13 mm [0.005 in]. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)

If surface flatness is **not** within limits, replace the exhaust manifold.

Note : A charge air cooler malfunction can cause progressive damage to the exhaust manifold. If the exhaust manifold is damaged, check the charge air cooler. Refer to Procedure 010-027 in Section 10.

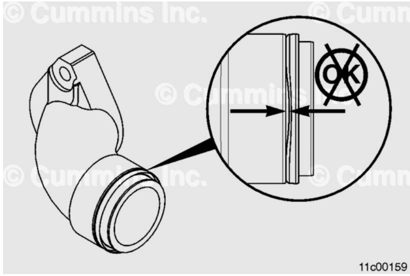
(/qs3/pubsys2/xml/en/procedures/377/377-010-027.html)

Inspect for cracks or other damage.



Check that the exhaust seal is fully seated against the shoulder of the exhaust manifold male end section.

If the exhaust seal ring is **not** fully seated against the shoulder, the exhaust seal ring **must** be replaced.



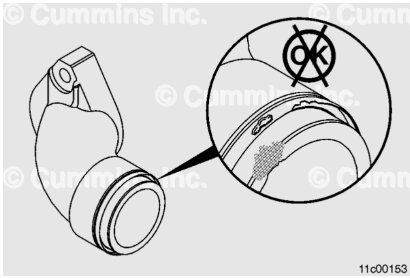
Inspect the exhaust manifold seal bore for wear.

Inspect the exhaust seal ring for signs of damage or excessive wear.

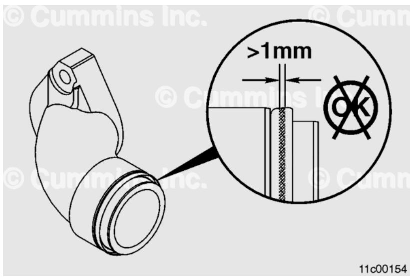
The exhaust seal ring **must not** show signs of dents or other signs of damage.

Measure the wear width on the exhaust seal ring.

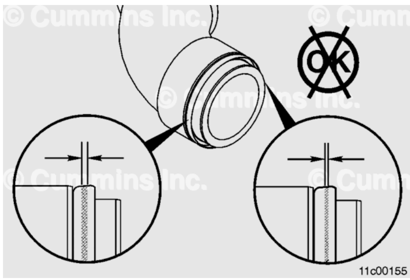
The wear width **must not** be more than 1.0 mm [0.04 in] wide anywhere on the seal.



A wear width larger than 1.0 mm [0.04 in] indicates excessive wear and the exhaust seal **must** be replaced.



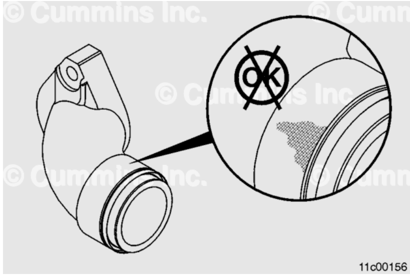
The wear width **must** be uniform around the seal ring. If one side of the exhaust seal ring is worn more than the other, the exhaust seal ring **must** be replaced.



Inspect the exhaust manifold slip-joints for signs of soot leakage.

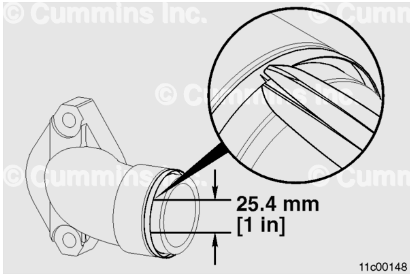
If there is no evidence of an exhaust leak from the slip-joints and there are no visible cracks or other damage, do **not** remove the exhaust manifold seal rings.

If there is a malfunction or other damage, remove the exhaust manifold seal rings.



Remove the exhaust seal ring, located within the slip-joint.

Use tin snips or side cuts to cut half way through the seal in two locations about 25.4 mm [1 in] apart.

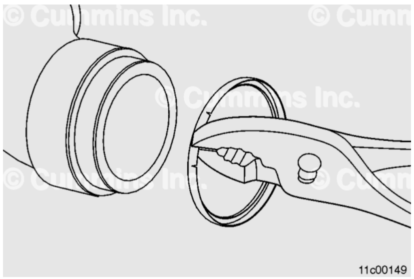


Note : Take care not to damage the seal carrier surface.

Grasp the portion of the seal between the cuts with a pair of pliers.

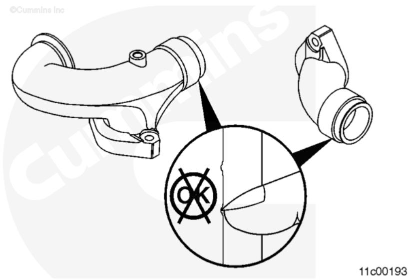
Twist and pull the seal off of the exhaust manifold end sections.

Discard the exhaust seal rings.

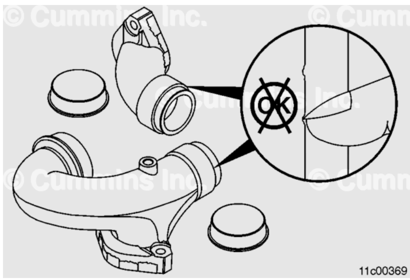


Inspect the exhaust seal ring mating surfaces of the male and female sections of the exhaust manifolds.

If scratches or nicks can be felt with a fingernail on the exhaust seal ring mating surfaces, the exhaust manifold section **must** be replaced.



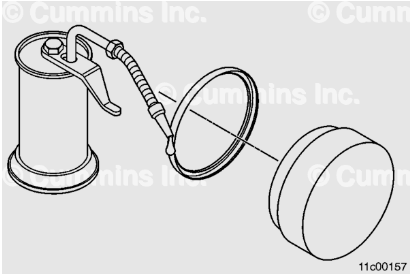
Cover the open points on the exhaust manifold with heavy tape to prevent engine contamination.



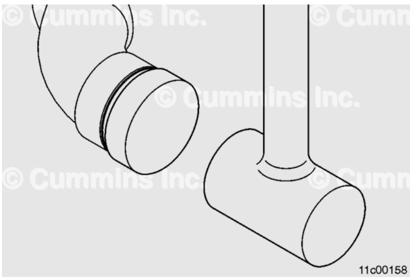
Assemble

Place the exhaust seal ring onto the lip of the exhaust seal installer, Part Number 4918709, or equivalent.

Use a light coat of grease or clean engine oil to lubricate the exhaust seal ring during assembly onto the male and female sections of the exhaust manifolds.

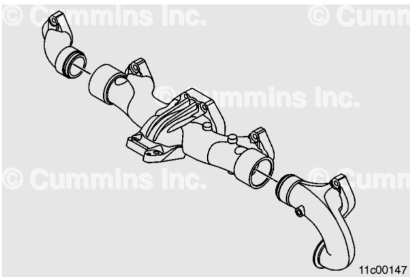


Position the seal installer and seal ring in line with the sealing surface on the exhaust manifold. Strike with a lead or dead blow hammer until the seal is fully seated.



Install the exhaust manifold end sections to the center section.

The end section bolt holes **must** be 152.4 mm [6 in] center to center before installing onto the cylinder head.



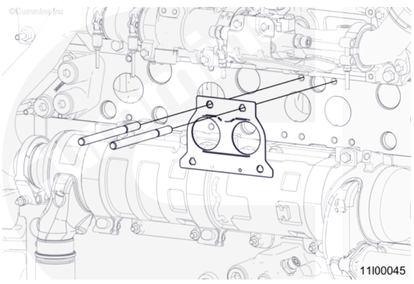
Install

Remove the tape from the open exhaust ports and cylinder head.

Install two guide studs into the top center section mounting location.

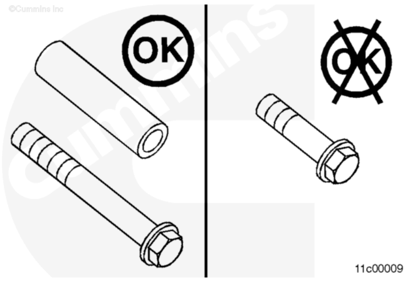
Do **not** use adhesive, grease, or heavy grease to hold the gaskets in place on the cylinder head.

Install a new center section gasket.



Special capscrews with spacers are required for this particular manifold.

Do **not** use short capscrews.



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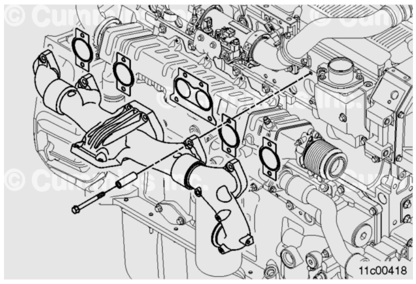


Install the exhaust manifold over the guide studs.

Apply a film of high-temperature anti-seize compound to the capscrew threads to make sure of proper loading on the capscrews.

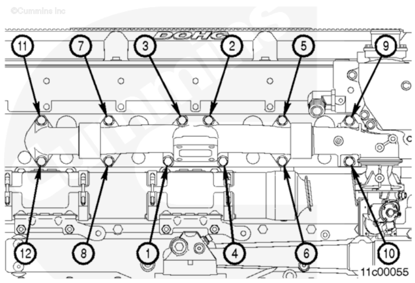
Slide each manifold gasket between the exhaust manifold and the cylinder head.

Install the capscrews and spacers through the exhaust manifold and gasket into the cylinder head.

Remove the two guide studs and install the two remaining capscrews.

Tighten the capscrews in the sequence shown.

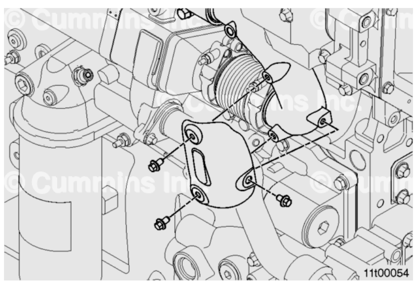
Torque Value: 60 n•m [44 ft-lb]



Apply high temperature anti-seize compound to the mounting capscrews.

Install the three heat shield mounting capscrews and the heat shield.

Torque Value: 23 n•m [204 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-027.html)
- Install the VGT. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)
- Remove the protective caps and heavy tape.
- Install the charge air cooler piping. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Connect the battieries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for air leaks and exhaust gas leaks.

Last Modified: 28-Aug-2019